

Module/Pathway/Taxon Review

Peptidoglycan precursor biosynthesis and lipid II export in *Pseudomonas putida* KT2440

- **Organism:** *Pseudomonas putida* KT2440 (PSEPK, NCBI taxon 160488, proteome UP0000000556)
 - **Module bucket:** kegg:ppu00550 ("Peptidoglycan biosynthesis"); module area *lipid_cell_envelope_metabolism*
 - **Module scope (as commissioned):** UDP-GlcNAc → UDP-MurNAc-pentapeptide → lipid I → lipid II → lipid II export. Ends at the flippase step; **excludes** glycan polymerization, transpeptidation/cross-linking, carrier recycling, and cell-wall remodeling.
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1. Executive summary

The peptidoglycan-precursor-and-lipid-II-export module is **fully satisfiable** in *P. putida* KT2440. All ten conceptual steps are encoded by identifiable, mostly single-copy genes with EC/domain evidence consistent with the expected model.

Two curation-critical points dominate this review:

1. **The terminal step (step 10, lipid II export) IS present but was OMITTED from the candidate list.** MurJ is encoded by **PP_0601 (Q88Q94)**, a MurJ/MviN-family protein (Pfam PF03023, InterPro IPR004268). It was excluded because the seeding bucket (KEGG map ppu00550) does not catalog the membrane flippase. Step 10 should be marked **covered**, not gap.
2. **The 23-gene candidate list is over-inclusive.** Only ~10 of 23 genes belong to this module. The other ~13 are downstream polymerases/transpeptidases/carboxypeptidases and undecaprenyl-carrier enzymes that the module scope explicitly excludes — a boundary artifact of using a whole KEGG-map bucket to seed a narrower biosynthetic module.

Evidence is largely **homology/annotation-based with strong within-Proteobacteria transfer**; there is little *KT2440-direct* genetic characterization of these specific loci, which is normal for essential housekeeping genes.

2. Target-organism pathway definition

Included process (KT2440): the cytoplasmic and inner-membrane-associated synthesis of the peptidoglycan precursor. Sequentially: enolpyruvyl transfer onto UDP-GlcNAc (MurA) → reduction to UDP-MurNAc (MurB) → sequential ligation of L-Ala, D-Glu, meso-DAP, and the D-Ala-D-Ala dipeptide onto UDP-MurNAc (MurC, MurD, MurE, MurF; D-Ala-D-Ala supplied by Ddl) → transfer of phospho-MurNAc-pentapeptide to the undecaprenyl-phosphate carrier to form **lipid I** (MraY) → GlcNAc addition to form **lipid II** (MurG) → translocation of lipid II to the periplasmic face (MurJ). KT2440 is a Gram-negative rod with **DAP-type (meso-A2pm) peptidoglycan**, consistent with the meso-DAP-adding MurE (EC 6.3.2.13) rather than a Lys-adding MurE.

Neighboring pathways/maps to keep SEPARATE: - **PG polymerization & cross-linking** (SEDS glycosyltransferases FtsW/RodA, class-A/B PBPs) — maps ppu01501/ppu01502; downstream of this module. - **Cell-wall remodeling** — D,D-carboxypeptidases (DacA/DacB). - **Undecaprenyl carrier metabolism** — Und-PP synthase (UppS, ppu00900) and Und-PP phosphatase/BacA (UppP, ppu00552); carrier supply/recycling, explicitly excluded. - **D-amino-acid supply** — alanine racemase, glutamate racemase (MurI), DAP epimerase (DapF); convergent inputs sitting in amino-acid metabolism maps.

Alternate names / database definitions: MraY = phospho-N-acetylmuramoyl-pentapeptide-transferase (translocase I). MurJ = MviN (historical). Ddl = D-Ala-D-Ala ligase (DdlA/DdlB). The *mur/mra* cluster is the **dcw (division and cell wall) cluster**. KEGG ppu00550 is broader than this module (it is the whole "Peptidoglycan biosynthesis" map, including polymerization and PBPs).

3. Expected step model and satisfiability call

#	Step	Expected player	KT2440 gene (locus / UniProt)	Call
1	Enolpyruvyl transfer to UDP-GlcNAc	MurA	murA / PP_0964 / Q88P88 (EC 2.5.1.7)	covered
2	UDP-MurNAc formation	MurB	murB / PP_1904 / Q88LM5 (EC 1.3.1.98)	covered
3	L-Ala addition	MurC	murC / PP_1338 / Q88N75 (EC 6.3.2.8)	covered
4	D-Glu addition	MurD	murD / PP_1335 / Q88N78 (EC 6.3.2.9)	covered
5	meso-DAP addition	MurE	murE / PP_1332 / Q88N81 (EC 6.3.2.13)	covered
6	D-Ala-D-Ala synthesis	Ddl	ddlB / PP_1339 / Q88N74 and ddlA / PP_4346 / Q88EV6 (EC 6.3.2.4)	covered (2 paralogs)
7	UDP-MurNAc-pentapeptide formation	MurF	murF / PP_1333 / Q88N80 (EC 6.3.2.10)	covered
8	Lipid I formation	MraY	mraY / PP_1334 / Q88N79 (EC 2.7.8.13)	covered
9	Lipid II formation	MurG	murG / PP_1337 / Q88N76 (EC 2.4.1.227)	covered
10	Lipid II export	MurJ	murJ / PP_0601 / Q88Q94 (MurJ/MviN family)	covered — <i>but was absent from candidate list</i>

Convergent inputs (present, but outside module boundary): glutamate racemase MurI (PP_0736, EC 5.1.1.3 → D-Glu for MurD); DAP epimerase DapF (PP_5228 and PP_3790, EC 5.1.1.7 → meso-DAP for MurE); alanine racemase activity from broad-specificity Alr (PP_3722, EC 5.1.1.10) and/or catabolic DadX (PP_5269, EC 5.1.1.1 → D-Ala for Ddl).

No step is **not_expected_in_target_taxon**; this module is universal in walled Gram-negative bacteria and KT2440 is no exception.

4. Candidate genes and evidence

4a. In-scope, high-confidence (module backbone)

- **MurA (PP_0964)**: EPSP-synthase family, MurA subfamily; single copy (no MurZ paralog). First committed, non-redundant step; fosfomycin target. Evidence: sequence/family + universal biochemistry. Transfer: strong.
- **MurB (PP_1904)**: UDP-N-acetylenolpyruvoylglucosamine reductase (EC 1.3.1.98). Single copy. Homology-based; strong.
- **MurC–MurF (PP_1338, PP_1335, PP_1332, PP_1333)**: the Mur ligase cascade, single copy each, clustered in the dcw locus (PP_1331–PP_1339). EC numbers match the expected chemistry exactly. MurE is the **meso-DAP-adding** form (correct for a DAP-type Gram-negative). Strong.
- **MraY (PP_1334)**: phospho-MurNAc-pentapeptide transferase (EC 2.7.8.13), integral-membrane translocase I. Strong.
- **MurG (PP_1337)**: lipid I β -GlcNAc-transferase (EC 2.4.1.227). Strong.
- **DdlA (PP_4346) & DdlB (PP_1339)**: two genuine D-Ala-D-Ala ligase paralogs. DdlB sits in the dcw cluster (likely primary biosynthetic ligase); DdlA is dispersed. Redundancy makes the step robust to single-gene loss. Caveat: physiological dominance of the two paralogs is not experimentally resolved in KT2440.

4b. In-scope terminal step, MISSING from candidates

- **MurJ (PP_0601 / Q88Q94)**: MurJ/MviN family (Pfam PF03023; InterPro IPR004268), ~512 aa, MOP-superfamily topology — the diagnostic signature of a lipid II flippase. It is the **sole** PF03023-family member in KT2440 (two orthogonal UniProt queries return only this entry), i.e. a **single-copy, non-redundant** step — unlike multi-paralog Firmicutes (*B. subtilis* has four MviN homologs that are individually dispensable,

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so no cryptic paralog offsets its omission. By transfer from essential single-copy MurJ in *E. coli* and *Burkholderia*, it is expected to be essential in KT2440. Not in the 23-gene metadata

because the KEGG map bucket omits the transporter. **This is the single most important correction to the candidate list.**

4c. Out-of-scope candidates (legitimate genes, wrong module)

Downstream PG synthesis/remodeling and carrier metabolism, to be reassigned: - **SEDS/PBP polymerases & transpeptidases:** pbpC (PP_0572), ftsI/PBP3 (PP_1331), ftsW (PP_1336), mrdB/RodA (PP_4806), mrdA-I (PP_3741), mrdA-II (PP_4807), mrcA/PBP1a (PP_5084), mrcB/PBP1b (PP_4683), mtgA (PP_5107). - **D,D-carboxypeptidase remodeling:** dacA (PP_4803), dacB (PP_2098). - **Undecaprenyl-carrier metabolism:** uppS (PP_1595, Und-PP synthase; carrier *supply*), uppP (PP_2862, Und-PP phosphatase/BacA; carrier *recycling*).

5. Gaps, ambiguities, and likely over-annotations

- **Apparent gap that is actually covered:** step 10 (export). Resolve by adding PP_0601. Do **not** attempt to satisfy step 10 with FtsW (PP_1336): FtsW was historically proposed as the flippase but is now established as a SEDS peptidoglycan glycosyltransferase; the flippase is MurJ (the "FtsW vs MurJ vs AmJ" debate is resolved in favor of MurJ for lipid II translocation).
- **Paralog ambiguity:** DdlA vs DdlB (step 6) — both real; dominance unresolved. Two DapF entries (PP_5228, PP_3790) — a possible epimerase paralog or an over-called annotation; both are *input-supply* enzymes, outside the module.
- **Broad EC/GO over-propagation:** EC 2.4.99.28 (peptidoglycan glycosyltransferase) is shared by MtgA, FtsW, RodA, PBP1a/1b — this shared EC drove several polymerases into the ppu00550 bucket. EC 3.4.16.4 (D,D-carboxypeptidase/transpeptidase) similarly pulled in FtsI, MrdA, DacA. These are annotation-map artifacts, not evidence of module membership.
- **Carrier boundary:** uppS/uppP border the module (Und-P is the acceptor for MraY) but the commissioned scope excludes carrier recycling. Keep them adjacent, not inside.
- **Evidence caveat:** all calls are homology/family-based; no KT2440-*direct* knockout/biochemical study was located for these specific loci. Transfer within Gram-negative Proteobacteria (*E. coli*/*Pseudomonas*/*Burkholderia*) is strong for the essential Mur enzymes and for MurJ.

6. Module and GO-curation recommendations

- **Mark covered:** steps 1–10, using the mapping in Section 3. In particular, record step 10 = **PP_0601 (MurJ)**.
 - **module_needs_revision (candidate list, not the module logic):** add PP_0601; remove/reassign the 13 out-of-scope downstream genes to the correct adjacent modules (PG polymerization/cross-linking; carrier metabolism; remodeling).
 - **candidate_uncertain:** none of the backbone steps; the only residual uncertainty is *which* Ddl paralog is primary (annotation-level, not satisfiability-level).
 - **not_expected_in_target_taxon:** none.
 - **New module documents / GO requests:** likely **not** needed for GO terms (MurJ, MraY, MurG, all Mur ligases have established GO/EC). A curation note should record that the seeding bucket (whole KEGG map) is broader than this module, so future bucket→module seeding for lipid-cell-envelope should (a) inject the transporter/flippase step manually and (b) exclude EC 2.4.99.28 / EC 3.4.16.4 polymerase/carboxypeptidase genes.
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7. Genes to promote to full fetch-gene review

1. **PP_0601 (murJ / Q88Q94)** — highest priority: newly identified terminal step, unreviewed TrEMBL ("Probable"), needs confidence upgrade and formal addition to the module.
 2. **PP_1339 (ddlB)** and **PP_4346 (ddlA)** — resolve paralog roles / which is essential.
 3. **PP_1334 (mraY)** and **PP_1337 (murG)** — membrane steps flanking export; confirm single-copy essentiality annotations.
 4. (Lower priority) **PP_0964 (murA)** — confirm fosfomycin-target/essentiality flag as the single non-redundant first step.
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8. Key references

- Ruiz N. (2008) *PNAS* 105:15553–15557. Bioinformatic identification of MviN/MurJ as the *E. coli* lipid II flippase.

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- Mohamed YF, Valvano MA. (2014) *Glycobiology/J. Bacteriol.* A *Burkholderia cenocepacia* MurJ (MviN) homolog is essential for PG synthesis and viability; depletion causes lipid II precursor accumulation.

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- Ruiz N. (2015) *Curr. Opin. Microbiol.*/review. Lipid flippases for bacterial PG biosynthesis; FtsW vs MurJ vs AmJ debate.

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- Fay A, Dworkin J. (2009) *J. Bacteriol.* 191:6020–6028. *B. subtilis* MviN/MurJ homologs are not essential — a caveat that MurJ essentiality is lineage-dependent (strong in Gram-negatives).

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- Molina-Henares MA et al. (2010) *Environ. Microbiol.* Genome-wide conditionally essential genes of *P. putida* KT2440 (context for KT2440 gene essentiality).

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- UniProt Knowledgebase, proteome UP000000556 (*P. putida* KT2440): entries Q88P88, Q88LM5, Q88N75/78/81/80/79/76, Q88N74, Q88EV6, Q88Q94, Q88PW2, plus racemase/epimerase inputs. Family/domain evidence: Pfam PF03023, InterPro IPR004268 (MurJ/MviN).

Evidence class summary: Steps 1–9 — direct family/EC assignment + universal biochemistry, strong within-Proteobacteria transfer. Step 10 — domain-level (MurJ/MviN, PF03023) assignment; function transferred by strong homology from *E. coli/Burkholderia* (no KT2440-direct experiment). Scope corrections — based on module definition + KEGG map structure + shared EC over-propagation.
